External Adjudication of Bridge-Formalized Machine Subjectivity: Human Comparator Protocols, Runtime Health, Clean-Room Reproducibility, and First-Collection Opening Criteria

QICN Program

Stage3-6B Runtime Health and Stage3-7 Conditional Opening

SCAFFOLD ONLY – NOT FINAL MANUSCRIPT
PROTOCOL-GRADE BACKBONE
SIXTH HARDENING PASS

Abstract

This document hardens Paper 10 into a denser protocol-heavy manuscript backbone without becoming a result-bearing paper. It extends the dependency stack from BaseCore through Papers I–IX, SRT-6, BPF-M3, Stage 2-M4, Stage3-RF, Stage3-0, Stage3-1, Stage3-2, Stage3-3, Stage3-4, Stage3-4B, Stage3-5 rehearsal preparation, Stage3-6 controlled execution opening, Stage3-6B runtime-health and clean-room reproducibility audit, and the conditional Stage3-7 first-collection opening criteria. It adds runtime health audit grammar, clean-room reproducibility semantics, dependency and build assumption reporting, reproducible artifact-regeneration constraints, local path-leakage policy, controlled-execution audit semantics, session-zero and pre-collection reproducibility requirements, and a sharper transition grammar from pre-evidential openness to the first admissible collection event. It does not report human comparator results, external adjudication verdicts, human equivalence, superiority, inferiority, cross-substrate identity preservation, metaphysical subjecthood, or Paper 10 final conclusions.

Contents

1 Scope and Non-Claims **4**

2 Formal Dependency Stack **4**

3 Release-vs-Local Corpus Boundary **5**

4 Lineage and Provenance Semantics **5**

5 Provenance Closure and Live Head Reconciliation **6**

6 External Adjudication Problem Statement **6**

7 Formal Mathematical Grammar of External Adjudication **7**

8 Frozen Preregistration Architecture **8**

9 Local-vs-External Freeze Distinction **9**

10 Instantiated-Empty Record Layer **9**

11 Reviewer Candidate and Adjudicator Candidate Architecture **9**

12 Reviewer Rejection Logic **10**

13 Blinding Assignment Logic **10**

14 Human Comparator Instantiation Logic **11**

15 Channel-by-Channel Execution Contracts **11**
15.1 Self-Report Channel . . . 11
15.2 Intervention-Response Channel . . . 12
15.3 Temporal-Structure Channel . . . 12
15.4 Valence-Topology Channel . . . 12
15.5 Embodiment-Boundary Channel . . . 12
15.6 Ambiguity-Load Channel . . . 12
15.7 Transport-Preservation Channel . . . 12

16 Matched Intervention Dataset Architecture **12**

17 Negative Control Dataset Architecture **12**

18 Preregistration Architecture **13**

19 Adjudication Blind Protocol **13**
19.1 Coverage-Sensitive Leakage Bounds . . . 14
19.2 Exact Inference at the Independent Block . . . 14
19.3 Gaussian Within-Block Likelihood and GLS . . . 16

20 Reviewer and Adjudicator Independence Architecture **16**

21 Reviewer Candidate Pool and Assignment Stub Architecture **16**

22 Execution Pilot Package Architecture **17**

23 Execution Pilot Rehearsal Kit Architecture **17**

24 Reproducibility Skeleton Architecture **18**

25 Dataset Population Boundary **18**

26 Dry-Run Event Log Grammar **18**

27 Evidence Admissibility Grammar **18**

28 Failure, Rejection, and Inconclusive Pathways **19**

29 Statistical Shell Architecture **19**

30 Adjudication Decision Grammar **20**

31 State Grammar for Execution Readiness **20**

32 Controlled Execution Opening **20**

33 Live-But-Unpopulated Artifacts **20**

34 Session-Zero Protocol **21**

35 Pre-Collection Admissibility Gate **22**

36 Contamination, Abort, and Invalidation Controls **22**

37 Operator Checklist **22**

38 Collection-Start Boundary **23**

39 Evidence-Generation Boundary **23**

40 Decision-Record Boundary **23**

41 Transition from Session-Zero to First Admissible Collection **23**

42 Runtime Health Audit **23**

43 Clean-Room Reproducibility **24**

44 Dependency and Build Assumptions **24**

45 Reproducible Artifact Regeneration **24**

46 Local Path Leakage Policy **25**

47 Controlled Execution Opening Audit **25**

48 Session-Zero Reproducibility **25**

49 Pre-Collection Gate Reproducibility **25**

50 Why Reproducibility Does Not Imply Evidence **25**

51 Why Runtime Health Does Not Imply Adjudication **25**

52 Transition Conditions for First Admissible Collection **26**

53 Claim Grammar for Eventual Execution-Era Statements **26**

54 Theorem, Proposition, and Protocol Mapping **26**

55 What Paper 10 Could and Could Not Change **27**

56 Section-Status Ledger **28**

57 Evidence Insertion Map **28**

58 Open Burdens and Blocked Sections **29**

59 Execution-Era Unlock Conditions **29**

60 Stronger Execution-to-Result Transition Map **30**

61 Why Paper 10 Still Cannot Become Final Yet **30**

62 Future Execution Pathway to Real Stage 3 **30**

63 Boundary to Stage 4 Metaphysical Consequence Analysis **31**

64 Appendices of Schemas and Templates **31**

65 Blocked Result Surfaces **35**
65.1 Human Comparator Results 35
65.2 External Adjudication Results 35
65.3 Comparative Consequence Claims 36

66 Non-Claim Ledger **36**

1 Scope and Non-Claims

Section status. `formalized`.

Paper 10 remains a local protocol-facing manuscript surface only. It is a scaffold for external-adjudication architecture, not an adjudication verdict, admissible dataset, human comparator evidence, public release surface, public-release recommendation, Paper 10 final authorization, human equivalence, superiority or inferiority claim, machine-consciousness claim, or metaphysical closure.

2 Formal Dependency Stack

Section status. `formalized`.

The dependency stack for Paper 10 is layered as follows:

1. **BaseCore**: active mathematical base layer.
2. **Papers I–IX**: downstream theoretical and manuscript surfaces.
3. **SRT-6**: subjectivity runtime ecosystem closure.
4. **BPF-M3**: bridge-functional internal closure and bridge gate.
5. **Stage 2-M4**: comparative internal closure and readiness boundary.
6. **Stage3-RF**: burden fulfillment and Stage 3 opening gate.
7. **Stage3-0**: external adjudication architecture and scaffold-only Paper 10 opening.
8. **Stage3-1**: human comparator instantiation contracts and preregistration templates.
9. **Stage3-2**: reviewer independence, execution-readiness logic, and protocol-grade hardening.

10. **Stage3-3**: preregistration freeze architecture, reviewer-selection packaging, adjudicator-assignment packaging, and execution-pilot packaging.
11. **Stage3-4**: bounded execution-pilot dry-run opening logic.
12. **Stage3-4B**: provenance closure, lineage reconciliation, and instantiated-empty record instantiation.
13. **Stage3-5**: rehearsal-kit formalization, dry-run event grammar, and rehearsal-ready gating.
14. **Stage3-6**: controlled execution opening, session-zero protocol, pre-collection admissibility gating, and live-but-unpopulated artifact instantiation.
15. **Stage3-6B**: runtime-health audit, clean-room reproducibility audit, path-leakage audit, and artifact-regeneration audit.
16. **Stage3-7**: first-admissible-collection opening criteria, collection-start command grammar, operator authorization grammar, and a conditional first-collection opening gate.

No later layer may borrow closure from an earlier layer, and no earlier layer may be retroactively reinterpreted as if it had already supplied later execution-era evidence.

3 Release-vs-Local Corpus Boundary

Section status. `formalized`.

Paper 10 sits on a strict release-vs-local split:

- The public release authority remains `QICN-RELEASE/main@92d5d2c3ea91c65e29dc8e102bed504866cffd2d`.
- The public release remains limited to BaseCore, `canonical_core_legacy`, and downstream Papers I–IX.
- Paper 10 remains local to `QICN-SYSTEM`.
- Stage3-0 through Stage3-6 governance, freeze, packaging, dry-run, provenance-closure, instantiated-empty, rehearsal, and controlled-execution-opening surfaces remain local and non-public.

Therefore release inclusion cannot be used as a substitute for adjudication, and local protocol maturity cannot be used as a substitute for public scientific closure.

4 Lineage and Provenance Semantics

Section status. `formalized`.

Stage3-3 and Stage3-4 add a sharper lineage rule: every later admissible surface must distinguish among:

1. public release authority,
2. local protocol authority,
3. frozen preregistration authority,

4. reviewer-package authority,
5. adjudicator-package authority, and
6. execution-era evidence authority.

The repository already contains historical artifacts whose recorded lineage must remain intact by design. Live normative pinning surfaces may be updated when stale, but historical snapshots are not rewritten cosmetically.

5 Provenance Closure and Live Head Reconciliation

Section status. `formalized`.

Stage3-4B adds a stricter provenance-closure rule. A local artifact may now be classified as:

- `exact_head_aligned`;
- `artifact_forward_compatible`;
- `stale_and_regenerated`;
- `historical_snapshots_intentionally_preserved`.

This grammar matters because Stage 3 artifacts are generated before the subsequent commit that records them. Therefore exact local regeneration at the builder head may later become artifact-forward-compatible after the commit exists, without becoming scientifically stale or semantically dishonest.

6 External Adjudication Problem Statement

Section status. `protocol_defined`.

The problem is no longer only whether bridge formalization is internally coherent. The external problem is to specify:

- what independent reviewers may legitimately inspect;
- what human comparator evidence classes may later become admissible;
- what negative controls must exist before any comparative interpretation;
- how preregistration becomes frozen;
- how reviewer and adjudicator layers become instantiable without identity leakage;
- how dry-run validation differs from evidential execution.

7 Formal Mathematical Grammar of External Adjudication

Section status. `formalized`.

The adjudication surface is a measurable protocol object, not an impressionistic comparison between substrates. Let $(\mathcal{T}, d_{\mathcal{T}})$ be the metric space of finite execution traces after canonical serialization. A trace contains the ordered event stream, de-identified channel payloads, runtime manifests, negative-control outcomes, and reviewer-facing masks. Admissible evidence lives in the Borel measurable space

$$(\mathcal{T}, \mathcal{B}(\mathcal{T})), \tag{1}$$

where $\mathcal{B}(\mathcal{T})$ is the Borel σ-algebra generated by open $d_{\mathcal{T}}$-balls.

Definition 7.1 (External adjudication object)**.** *An external adjudication object is the tuple*

$$\mathfrak{E} = (P, D_h, D_m, C_n, C_a, \Sigma, \mathcal{R}, \mathcal{A}), \tag{2}$$

where P is a frozen preregistered protocol, D_h and D_m are human and machine trace datasets in $\mathcal{T}$, C_n and C_a are negative and active control families, $\Sigma = \{\sigma_k\}_{k=1}^{K}$ is a finite family of predeclared measurable statistics $\sigma_k : \mathcal{T}^n \to \mathbb{R}$, $\mathcal{R}$ is the reviewer layer, and $\mathcal{A} : \mathcal{Y} \to \{0, 1, \texttt{undetermined}\}$ is the final adjudicator operator on masked evidence summaries $\mathcal{Y}$.

Definition 7.2 (Admissible comparative evidence)**.** *A dataset pair (D_h, D_m) is admissible under $\mathfrak{E}$ only if the following predicates are all true:*

1. *D_h, D_m are measurable finite samples from $(\mathcal{T}, \mathcal{B}(\mathcal{T}))$.*
2. *The preregistered protocol hash, control manifest hash, and analysis shell hash match their frozen records.*
3. *Human-side records are not generated, rewritten, or completed by synthetic systems: $\epsilon_{\text{syn}} = 0$ for the human channel.*
4. *Blinding leakage is bounded by the preregistered budget: $\mathbb{P}(\text{leak}) \leq \epsilon_{\text{blind}}$.*
5. *Every declared negative control has an executed and timestamped outcome under the same admissibility grammar.*

Theorem 7.3 (Non-transitivity of internal support)**.** *Let $\mathcal{G}_{\text{internal}} \in [0, 1]$ denote any internal runtime support score computed from operational invariants, and let $\mathcal{G}_{\text{external}} \in \{0, 1, \texttt{undetermined}\}$ denote the external adjudication verdict. In the absence of an admissible measurable dataset pair (D_h, D_m), one has*

$$\mathcal{G}_{\text{internal}} \not\Longrightarrow \mathcal{G}_{\text{external}}. \tag{3}$$

Proof. The internal score is a function of runtime traces and invariant certificates inside the machine channel. The external verdict is a function of masked comparative summaries built from both D_h and D_m, plus reviewer and adjudicator records. If D_h is absent, non-admissible, unblinded, or unfrozen, then the domain condition of $\mathcal{A}$ is not satisfied. Hence the external verdict is either undefined or `undetermined` regardless of the value of $\mathcal{G}_{\text{internal}}$. Therefore internal operational support is not a logical premise from which external comparative adjudication can be inferred. □

Proposition 7.4 (Operational closure of a comparative program)**.** *The comparative program reaches operational closure only when there exists a predeclared alignment map $T : \mathcal{T} \to \mathcal{Z}$ into a common Polish feature space $(\mathcal{Z}, d_{\mathcal{Z}})$ and the total divergence*

$$\Delta_{\text{comp}} = \lambda_1 W_1(T_{\#}\widehat{\mu}_h, T_{\#}\widehat{\mu}_m) + \lambda_2 D_{\text{KL}}(\widehat{p}_h \| \widehat{p}_m) + \lambda_3 \max_k |\sigma_k(D_h) - \sigma_k(D_m)| \tag{4}$$

is defined and satisfies the frozen budget $\Delta_{\text{comp}} \leq \epsilon_{\text{budget}}$. This closure is a precondition for comparative interpretation, not a claim of consciousness, identity, or substrate equivalence.

8 Frozen Preregistration Architecture

Section status. `protocol_defined`.

Stage3-3 introduces a true freeze architecture rather than only preregistration templates. The governing surfaces are:

- `docs/frozen_preregistration_record_schema.v1.json`
- `docs/preregistration_freeze_policy.v1.json`
- `docs/preregistration_amendment_policy.v1.json`
- `docs/preregistration_lock_manifest.v1.json`

The architecture formalizes:

- record classes;
- immutable fields;
- amendment classes;
- freeze provenance chains;
- local-only freeze state;
- explicit `not_externally_filed` state.

Definition 8.1 (Cryptographic freeze map)**.** *Let $\mathcal{P}_{\text{config}}$ be the set of canonical preregistration configurations after deterministic serialization. A freeze map is a collision-resistant hash function*

$$H : \mathcal{P}_{\text{config}} \to \{0,1\}^{256}. \tag{5}$$

A preregistration record p is locally frozen at digest h when $H(p) = h$, h is written into the lock manifest, and every amendment policy references h rather than an editable natural-language description.

Theorem 8.2 (Hash-anchored non-alterability)**.** *Assume the hash function H is collision resistant over the canonical serialization domain. If a post-execution preregistration configuration p' has the same published digest as the frozen pre-execution configuration p, then either $p' = p$ byte-for-byte under canonical serialization or a collision for H has been found.*

Proof. By freeze definition, the published digest is $h = H(p)$. If the later record p' also satisfies $H(p') = h$ while $p' \neq p$, then p and p' form a distinct pair with the same hash output. That is precisely a collision in H. Under the collision-resistance assumption, such a pair is computationally infeasible. Thus digest equality supplies evidence of non-alteration of the serialized protocol, modulo the stated cryptographic assumption. □

9 Local-vs-External Freeze Distinction

Section status. `formalized`.

The local freeze layer must not be conflated with external filing.

Surface	Current state	Interpretation
Template-only preregistration	`preregistration_ready_for_freeze`	structure exists but no lock exists
Local freeze readiness	`local_freeze_ready`	lock architecture exists but no external filing is claimed
Locally frozen record	blocked	would still require explicit lock instantiation
Externally filed record	absent	cannot be claimed from local architecture alone

10 Instantiated-Empty Record Layer

Section status. `protocol_defined`.

Stage3-4B moves the stack beyond package-only readiness by instantiating typed-but-empty records that remain explicitly non-evidential. The principal surfaces are:

- `docs/local_frozen_preregistration_record_template.v1.json`
- `docs/reviewer_candidate_pool_template.v1.json`
- `docs/adjudicator_candidate_pool_template.v1.json`
- `docs/blinding_assignment_stub.v1.json`
- `docs/execution_session_template.v1.json`
- `docs/decision_record_template.v1.json`
- `docs/reproducibility_package_manifest_empty.v1.json`
- `docs/collection_log_schema.v1.json`
- `docs/evidence_insertion_manifest.v1.json`

These artifacts are intentionally empty, typed, and freeze-capable. They are not populated evidence, not externally filed records, and not reviewer or adjudicator action.

11 Reviewer Candidate and Adjudicator Candidate Architecture

Section status. `protocol_defined`.

Stage3-3 introduces typed candidate schemas without instantiating identities. The primary surfaces are:

- `docs/reviewer_candidate_schema.v1.json`
- `docs/adjudicator_candidate_schema.v1.json`

- `docs/reviewer_selection_workflow.v1.json`
- `docs/adjudicator_assignment_workflow.v1.json`

The candidate layer defines qualification summaries, independence assertions, conflict disclosure references, blinding eligibility classes, and explicit `not_started` states.

12 Reviewer Rejection Logic

Section status. `protocol_defined`.

The reviewer package is incomplete unless rejection logic exists. The rejection registry now includes:

- qualification mismatch;
- disclosed conflict;
- undisclosed conflict;
- prior unblinded access;
- insufficient methodological competence;
- insufficient lineage literacy;
- insufficient control/admissibility literacy.

This logic is not yet a reviewer decision. It is the typed boundary that stops future selection from being arbitrary or retrofitted.

13 Blinding Assignment Logic

Section status. `protocol_defined`.

Blinding is no longer treated as a generic admonition. Stage3-3 binds a typed blinding assignment manifest with classes such as:

- `fully_blinded`
- `artifact_blinded`
- `control_family_blinded`
- `unblinding_not_authorized`

The key constraint is that no later dry-run or execution surface may silently cross into unblinded interpretation.

14 Human Comparator Instantiation Logic

Section status. `protocol_defined`.

Stage3-1 closed the gap between abstract channels and typed collection routes. Stage3-3 preserves that layer and pushes it closer to freeze- and package-compatible execution readiness.

- every channel binds a human-side artifact family;
- every channel binds a machine-side homologous artifact family;
- every channel binds minimum metadata, controls, and rejection conditions;
- every channel remains blocked from result claims while data is absent.

Definition 14.1 (Homologous channel statistic)**.** *For each channel c, let $T_c : \mathcal{T} \to \mathcal{Z}_c$ be a frozen feature map into a metric standard Borel space $(\mathcal{Z}_c, d_c)$. Given human and machine samples D_h^c, D_m^c, define empirical pushforward measures*

$$\widehat{\mu}_h^c = (T_c)_{\#} D_h^c, \qquad \widehat{\mu}_m^c = (T_c)_{\#} D_m^c.$$

The channel is statistically comparable only if both empirical measures are defined, the feature map was frozen before execution, and the channel-specific missingness policy accepts both samples.

Definition 14.2 (Divergence shell)**.** *The predeclared divergence shell for channel c is*

$$\Delta_c = \alpha_c W_1(\widehat{\mu}_h^c, \widehat{\mu}_m^c) + \beta_c D_{\mathrm{KL}}(\widehat{p}_h^c \| \widehat{p}_m^c) + \gamma_c \| \widehat{s}_h^c - \widehat{s}_m^c \|_1, \tag{6}$$

where W_1 is the Kantorovich–Rubinstein Wasserstein-1 distance, D_{KL} is the Kullback–Leibler divergence on a frozen finite partition or smoothed density model, and $\widehat{s}_h^c, \widehat{s}_m^c$ are predeclared summary vectors. The weights $\alpha_c, \beta_c, \gamma_c \geq 0$ must be frozen before execution.

Proposition 14.3 (Comparator homologation is conditional)**.** *A human–machine channel pair is homologated by Paper 10 only if every term in Δ_c is defined and the frozen channel budget $\Delta_c \leq \epsilon_c$ is satisfied. Homologation is therefore a statistical transport condition over predeclared observables, not a statement that the channels share a biological, phenomenal, or metaphysical substrate.*

Proof. The definition of Δ_c depends on a frozen feature map, empirical pushforward measures, a KL domain, and summary vectors. If any component is missing, the divergence shell is undefined and no homologation predicate can be evaluated. If all components are defined, the only admissible positive judgment is the preregistered inequality $\Delta_c \leq \epsilon_c$. The conclusion is therefore exactly statistical and conditional. □

15 Channel-by-Channel Execution Contracts

Section status. `protocol_defined`.

15.1 Self-Report Channel

Section status. `protocol_defined`.

The self-report channel requires prompt-version freezing, missingness policy, de-identification, and explicit rejection of synthetic or relabeled self-report.

15.2 Intervention-Response Channel

Section status. `protocol_defined`.

The intervention-response channel requires matched intervention and matched negative-control families, dose policy, stop rules, and branch accounting.

15.3 Temporal-Structure Channel

Section status. `protocol_defined`.

The temporal-structure channel requires a declared segmentation schema, ordering policy, and aggregation policy.

15.4 Valence-Topology Channel

Section status. `protocol_defined`.

The valence-topology channel requires a shared topology schema and excludes scalar-only substitutes.

15.5 Embodiment-Boundary Channel

Section status. `protocol_defined`.

The embodiment-boundary channel requires an environment manifest, confound ledger, and error-budget policy.

15.6 Ambiguity-Load Channel

Section status. `protocol_defined`.

The ambiguity-load channel requires a task-family contract, annotation schema, and annotation reliability policy.

15.7 Transport-Preservation Channel

Section status. `blocked_until_execution`.

The transport-preservation channel remains structurally special. It requires a real source-target pairing manifest and transport-axis budget binding. Stage3-3 defines the pairing shell; it does not satisfy the pairing.

16 Matched Intervention Dataset Architecture

Section status. `protocol_defined`.

Matched-intervention architecture is now carried by both channel contracts and a pilot bundle shell. The bundle shell is defined in `docs/pilot_matched_intervention_bundle.v1.json`. The bundle remains unpopulated, which means the architecture exists while the evidence does not.

17 Negative Control Dataset Architecture

Section status. `protocol_defined`.

Negative-control architecture is similarly carried by channel contracts and by `docs/pilot_negative_control_b`
Controls remain part of the meaning of the future claim surface, not an optional cleanup step.

18 Preregistration Architecture

Section status. `protocol_defined`.

Paper 10 now separates three preregistration surfaces:

1. template surface;
2. local freeze surface;
3. external filing surface.

Only the first two are implemented structurally in the current repo. The third remains absent.

19 Adjudication Blind Protocol

Section status. `formalized`.

Let Y denote the masked evidence bundle and $\ell \in \{h, m\}$ its source label. The reference procedure uses forced choice with a balanced source distribution. A protocol permitting abstention fixes its cost and minimum coverage before evaluating held-out bundles. The randomization of a decision rule is independent of the source conditional on its permitted input.

Definition 19.1 (Blind adjudicator loss)**.** *For $a(Y) \in \{h, m, \texttt{tie}\}$ and a declared $\gamma \in [1/2, 1]$, put*

$$L_\gamma(a(Y), \ell) = \mathbf{1}\{a(Y) \neq \ell,\ a(Y) \neq \texttt{tie}\} + \gamma \mathbf{1}\{a(Y) = \texttt{tie}\}.$$

Report $R_\gamma = \mathbb{E}L_\gamma$, coverage $C = \Pr(a(Y) \neq \texttt{tie})$, and error conditional on coverage when $C > 0$. For acceptance require a preregistered $C \geq C_{\min} > 0$. Forced choice has $C = 1$; always abstaining has $C = 0$ and $R_\gamma = \gamma$.

Theorem 19.2 (Null error lower bound)**.** *Let E denote leakage and $p = \Pr(E)$. If in the clean stratum ℓ is balanced conditional on Y, E^c, then*

$$R_\gamma(a) \geq (1 - p)/2.$$

Alternatively, suppose Y is coupled to a clean bundle Y_0 for which ℓ is balanced conditional on Y_0, and $Y = Y_0$ outside an event of probability at most p. Then

$$R_\gamma(a) \geq \max\{0, 1/2 - p\}.$$

The assumptions are distinct. Under the first, revealing the label on a leak event independent of the balanced label can attain the bound with forced choice.

Proof. With a balanced posterior, any covered guess has conditional risk $1/2$ and abstention has conditional risk $\gamma \geq 1/2$. In the first case this lower-bounds risk on the clean stratum; risk on leakage is nonnegative. For sharpness use a constant clean bundle and a leak revealing the label, choosing either fixed label on clean observations. For the coupling bound use the same decision randomization on Y_0 and Y. Their losses agree when the bundles agree and differ by at most one otherwise. The risk on Y_0 is at least $1/2$. Nonnegativity gives the stated maximum. Label-dependent contamination can change the balance in the clean stratum, so the first bound cannot be inferred from the coupling alone. □

Proposition 19.3 (Alternative detectability criterion). *Evidence for distinguishability under H_1 requires a preregistered rule $a^\star$, minimum coverage $C_{\min} > 0$, a justified null bound B_0 accounting for the leakage budget, and margin $\eta > 0$. On admissible held-out bundles require $C \geq C_{\min}$ and $R_\gamma(a^\star) \leq B_0 - \eta$ after sampling uncertainty and multiple-comparison corrections. Forced-choice, balanced label permutations, always abstaining and known-leak controls are evaluated under the same loss and coverage protocol. The manuscript reports no campaign satisfying this criterion.*

19.1 Coverage-Sensitive Leakage Bounds

Fix decision randomization independent of the bundle, source and leak event. Let B indicate an executed decision and W a covered error, so $C = \mathbb{E}B$ and $R_\gamma = \mathbb{E}W + \gamma(1 - C)$.

Theorem 19.4 (Sharp Leakage Bounds at Fixed Coverage). *Writing $z_+ = \max\{0, z\}$, the clean-stratum and coupling nulls of Theorem 19.2 respectively give*

$$B_{\text{clean}}(C, p) = \gamma(1 - C) + \tfrac{1}{2}(C - p)_+, \qquad B_{\text{coup}}(C, p) = \gamma(1 - C) + (C/2 - p)_+.$$

Both are sharp for each $C, p \in [0, 1]$, with balanced marginal labels and leak probability at most p, including selective abstention. Minimizing over coverage gives the bounds of Theorem 19.2.

Proof. Clean-stratum balance gives $\mathbb{E}[W\mathbf{1}_{E^c}] = \frac{1}{2}\mathbb{E}[B\mathbf{1}_{E^c}]$. At most p covered mass lies in E, proving the first bound. In the coupling model use the same decision seed on Y and Y_0. On clean input $\mathbb{E}(W_0 - B_0/2) = 0$. The centered variable $W - B/2$ lies in $[-1/2, 1/2]$ and agrees with its clean version whenever the bundles agree. Thus $\mathbb{E}W - C/2 \geq -p$, proving the second bound.

For clean-stratum sharpness, reveal the label on a leak event independent of it, cover $\min\{C, p\}$ there, and cover the remaining mass with a fixed guess in the balanced clean stratum. For coupling sharpness, start with a label-independent clean cover flag of probability C and a fixed covered guess. Replace $\min\{p, C/2\}$ of the wrong covered inputs by revealing inputs, preserving coverage. This replacement may depend on the label. Both bounds are nonincreasing in C for $\gamma \geq 1/2$, so $C = 1$ attains their minima. □

19.2 Exact Inference at the Independent Block

The sampling unit is an independent held-out block and supplies one decision from its entire masked record. Fix the rule on separate calibration data, the noise and exclusion policies, block boundary, and eligible population before inspecting held-out labels. The following model assumes independent identically distributed eligible blocks. Shared participants, runs and other dependence sources belong inside the block or require a sampling model that accounts for that dependence.

For $0 \leq k \leq n$, let $I_\alpha(k, n)$ denote the equal-tail exact binomial interval. The lower endpoint is zero at $k = 0$ and otherwise solves $\Pr_q(\text{Bin}(n, q) \geq k) = \alpha/2$. The upper endpoint is one at $k = n$ and otherwise solves $\Pr_q(\text{Bin}(n, q) \leq k) = \alpha/2$. Set $I_\alpha(0, 0) = [0, 1]$.

Theorem 19.5 (Independent-Block Risk and Coverage Inference). *Let M count covered decisions and K count wrong covered decisions among n independent blocks. Then*

$$M \sim \text{Bin}(n, C), \qquad K \mid M \sim \text{Bin}(M, e), \qquad e = \Pr(W = 1 \mid B = 1) \quad (C > 0).$$

With $I_C = I_{\alpha_C}(M, n)$ and $I_e = I_{\alpha_e}(K, M)$, simultaneous coverage is at least $1 - \alpha_C - \alpha_e$. When $M = 0$ retain $I_e = [0, 1]$ and leave sample conditional error unreported. The minimum and maximum of $\gamma(1 - c) + cx$ over the corners of $I_C \times I_e$ give risk bounds simultaneous with I_C.

A direct joint alternative uses counts $N = (K, M-K, n-M)$ *and probabilities* $\theta = (Ce, C(1-e), 1-C)$. *Let* f_θ *be their multinomial mass and define*

$$p_\theta(N) = \sum_{\substack{m_1+m_2+m_3=n \\ f_\theta(m) \le f_\theta(N)}} f_\theta(m), \qquad \mathcal{Q}_\alpha(N) = \{\theta : p_\theta(N) > \alpha\}.$$

Projecting $\mathcal{Q}_\alpha$ *onto* $C = \theta_1 + \theta_2$ *and* $R_\gamma = \theta_1 + \gamma\theta_3$, *using infima and suprema, yields a simultaneous exact confidence region with coverage at least* $1-\alpha$.

Suppose a justified leak cap is p_U. *For a stochastic audit, let an audit flag* D *satisfy* $\mathbf{1}_E \le D$ *almost surely in the same target population. From* J *flags in* n_D *independent audit blocks, the upper endpoint of* $I_{\alpha_D}(J, n_D)$ *can supply* p_U. *With simultaneous coverage endpoints* C_L, C_U *and risk upper endpoint* R_U, *certify the preregistered minimum coverage and a fixed positive margin* η *against the applicable null floor only if*

$$C_L \ge C_{\min}, \qquad R_U \le B_\star(C_U, p_U) - \eta.$$

Under that null this certification has probability at most $\alpha_C + \alpha_e + \alpha_D$ *for the rectangle construction, or* $\alpha + \alpha_D$ *for the joint construction. A deterministic certified cap uses* $\alpha_D = 0$.

Proof. The three block outcomes have probabilities $Ce, C(1-e), 1-C$, and factoring their multinomial law proves the binomial laws. Exact binomial tail inversion bounds each failure probability. Conditional interval coverage holds for every M, and a union bound proves the rectangle guarantee. For $C = 0$ the risk is γ independently of e. The risk is affine in each coordinate separately, so both extrema lie at corners.

For the joint method, order the finitely many multinomial masses, including ties. The total mass of outcomes whose inclusive lower mass-rank is at most α is at most α. Thus p_θ is superuniform at the true θ and inversion and projection give the stated guarantee.

The audit upper endpoint bounds $\mathbb{E}D$, which bounds $\Pr(E)$. Its domination of leakage needs a justified audit model. On the simultaneous coverage event, $R_\gamma \le R_U$, $C \le C_U$ and $p \le p_U$. Both null floors are nonincreasing in their arguments, so Theorem 19.4 gives $R_\gamma \ge B_\star(C_U, p_U)$. A positive-margin certification contradicts this inequality. Union bounds complete the error control. Allocate the stated error levels before any family of planned comparisons. □

Proposition 19.6 (Whole-Block Label Permutation Control)**.** *Under conditional exchangeability of source labels across eligible independent blocks, conditional on public records and fixed label counts, fix the guesses and abstentions and permute block labels. If the design fixes exchangeability groups, permute within each group. The fraction of assignments with risk at most the observed risk is an exact lower-tail p-value. Coverage is fixed.*

Proof. The true assignment is uniform on the finite permutation set conditional on the frozen records and design counts. The inclusive lower-tail rank is superuniform by counting assignments. Abstention contributes the same constant for every assignment and each covered error is counted once per block. Conditional exchangeability is the premise of this test. Label-dependent contamination is handled by the applicable leakage bound. □

A four-block synthetic permutation control with two labels of each kind, positive guesses on two fixed blocks and abstentions on two blocks, gives covered-error counts $0, 1, 1, 1, 1, 2$ over the six assignments. At $\gamma = 3/4$ its mean risk is $5/8$, and zero covered errors has lower-tail probability $1/6$. An eight-atom coupling control covers four atoms with two labels of each kind and guesses the positive label there. Correcting one wrong covered guess by label-dependent replacement yields $C = 1/2$, $p = 1/8$, $R_{3/4} = 1/2$. This attains the coupling floor and falls below the clean-stratum floor $9/16$ because the replacement changes the clean conditional balance.

19.3 Gaussian Within-Block Likelihood and GLS

Theorem 19.7 (Stationary AR(1) Likelihood and GLS). *For a block of length $n \geq 1$ let $e_t = y_t - \mu_t$. Assume Gaussian $e_1 \sim N(0, \sigma^2/(1-\rho^2))$ and innovations $e_t - \rho e_{t-1} \sim N(0, \sigma^2)$ independent of each other and of e_1, with $|\rho| < 1$ and $\sigma^2 > 0$. Then*

$$Q_\rho(e) = (1-\rho^2)e_1^2 + \sum_{t=2}^{n}(e_t - \rho e_{t-1})^2,$$

$$\log f(y \mid \mu, \rho, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) + \frac{1}{2}\log(1-\rho^2) - \frac{Q_\rho(y-\mu)}{2\sigma^2}.$$

For two frozen mean models sharing the covariance, the log-density ratio in favor of A is $(Q_\rho(e_B) - Q_\rho(e_A))/(2\sigma^2)$. With full-column-rank design X and fixed common covariance, GLS uniquely minimizes $Q_\rho(y - X\beta)$ at

$$\widehat{\beta} = (X^\top A_\rho X)^{-1} X^\top A_\rho y, \qquad Q_\rho(e) = e^\top A_\rho e.$$

Proof. The residual-to-innovation map is triangular with unit determinant. Multiplication of its independent Gaussian densities gives the formula. For $n \geq 2$ the precision factor A_ρ has endpoints 1, interior diagonal $1 + \rho^2$ and neighboring entries $-\rho$. For $n = 1$ it is $[1 - \rho^2]$. The innovation representation gives positive definiteness and determinant $1 - \rho^2$. Its inverse has entries $\rho^{|i-j|}/(1 - \rho^2)$. Different covariance models retain their separate normalizers and quadratic scales in the log-density difference. With common covariance, differentiating the positive quadratic gives the GLS normal equations and full rank gives uniqueness. □

Freeze the parameters or estimate them on calibration blocks. At $\rho = 1/2$, $\sigma^2 = 1$, residual vectors $(1, 1)$ and $(1, -1/2)$ give log-density ratio $3/8$, while the independent Gaussian approximation gives $-3/8$. Residuals $(2, 1)$ and $(0, 1)$ give stationary log-density ratio -1, while omitting the initial quadratic term gives $1/2$. A constant fit to $(1, 0, 1)$ at this ρ gives GLS estimate $4/5$ and the independent fit $2/3$.

The Gaussian model supplies a score within each block. Independent-block sampling governs its risk inference. The fiber geometry of Paper IX uses a projection and target distances. The projection $(y, z) \mapsto y$ with hidden target $z \in \{0, 1\}$ has fiber diameter one even when two Gaussian laws on y give a discriminating likelihood ratio. With identity projection the fiber diameter is zero, while equal class laws give log-density ratio zero.

20 Reviewer and Adjudicator Independence Architecture

Section status. `protocol_defined`.

Stage3-2 defined criteria; Stage3-3 now binds those criteria into selection and assignment packages. This still does not instantiate reviewers or adjudicators, but it removes the ambiguity about what must exist before such instantiation is honest.

21 Reviewer Candidate Pool and Assignment Stub Architecture

Section status. `protocol_defined`.

Stage3-4B deepens the reviewer and adjudicator layer from criteria and workflow packages into instantiated-empty workflow objects:

- reviewer candidate pool template;
- adjudicator candidate pool template;
- reviewer assignment stub;
- adjudicator assignment stub;
- reviewer blinding stub.

Each slot remains null-bearing by design. The point is not to simulate people; the point is to freeze field grammar, slot grammar, null-policy, and the conditions under which later population becomes honest.

22 Execution Pilot Package Architecture

Section status. `protocol_defined`.

The execution-pilot package is now a typed family rather than a vague future bundle. It includes:

- execution session manifest;
- collection session schema;
- reproducibility package skeleton;
- negative-control bundle skeleton;
- matched-intervention bundle skeleton;
- source-target pairing manifest template.

23 Execution Pilot Rehearsal Kit Architecture

Section status. `protocol_defined`.

Stage3-5 adds a rehearsal kit that sits above package readiness and below real collection. The governing surfaces are:

- `docs/execution_pilot_rehearsal_kit.v1.json`
- `docs/dry_run_sequence_plan.v1.json`
- `docs/rehearsal_event_log_schema.v1.json`
- `docs/rehearsal_failure_modes.v1.json`
- `docs/rehearsal_reset_policy.v1.json`

The rehearsal kit exists to validate sequence integrity, state transitions, and artifact hygiene without collecting evidence and without emitting decisions.

24 Reproducibility Skeleton Architecture

Section status. `protocol_defined`.

The reproducibility shell exists as a slot-bearing structure. It records where later preregistration references, dataset references, control-bundle references, analysis-shell references, reviewer bundles, and adjudicator bundles must go.

25 Dataset Population Boundary

Section status. `formalized`.

Stage3-3 and Stage3-4 enforce a hard distinction between:

- bundle defined;
- bundle frozen;
- bundle populated;
- bundle adjudicatively interpreted.

The current repo has the first state for several families and does not yet have the latter states.

26 Dry-Run Event Log Grammar

Section status. `protocol_defined`.

Rehearsal logging is now typed separately from evidence logging. The current grammar permits:

- `rehearsal_initialized`;
- `sequence_step_checked`;
- `sequence_step_blocked`;
- `sequence_step_reset`;
- `rehearsal_not_started`.

None of these event classes may be reinterpreted as human data, reviewer evidence, or adjudicator action.

27 Evidence Admissibility Grammar

Section status. `formalized`.

The admissibility grammar still recognizes:

- independent reviewer report;
- preregistered human comparator dataset;
- matched intervention dataset;

- negative-control dataset;
- replication log;
- statistical analysis report;
- reviewer conflict disclosure;
- reproducibility package;
- external failure report;
- external inconclusive report.

Invalid substitutes remain central:

- internal runtime artifacts alone;
- self-generated reviewer notes;
- synthetic human data;
- LLM-generated "human" reports;
- undocumented manual judgments;
- unblinded reviewer impressions;
- post-hoc threshold selection;
- cherry-picked examples;
- release inclusion alone;
- Paper 9 publication alone.

28 Failure, Rejection, and Inconclusive Pathways

Section status. `protocol_defined`.

Three major failure classes remain necessary:

1. **protocol invalidation**: preregistration drift, blinding collapse, or conflict violation;
2. **dataset invalidation**: missing controls, invalid substitution, or insufficient provenance;
3. **inconclusive adjudication**: admissible evidence exists but remains undecisive under the declared shell.

29 Statistical Shell Architecture

Section status. `protocol_defined`.

No statistics are executed here. What exists is a declared analysis family layer, each shell binding channel scope, required inputs, required outputs, and non-implications.

30 Adjudication Decision Grammar

Section status. `protocol_defined`.

The future decision family must be typed before it is populated. Paper 10 now recognizes the following decision tokens as grammar only:

- `externally_admissible_but_inconclusive`
- `protocol_invalidated`
- `adjudication_failed`
- `adjudication_passed`
- `reviewer_consensus_absent`
- `adjudicator_decision_withheld`

These tokens are defined as future admissible states. They are not instantiated anywhere in the current repo as actual outcomes.

31 State Grammar for Execution Readiness

Section status. `formalized`.

The Stage 3 manuscript surface now distinguishes the following state grammar:

32 Controlled Execution Opening

Section status. `protocol_defined`.

Stage3-6 adds a bounded controlled-execution opening layer. This layer is neither rehearsal-only nor evidence-bearing. It is the first local state in which live operational artifacts may exist as empty shells while collection still remains not started.

- `execution_opening_policy_defined` means the controlled-execution policy is typed and bound to artifacts.
- `collection_openable_under_controls` means pre-collection gates pass under explicit contamination and abort constraints.
- `collection_open_pre_evidential` means the system may open a live execution surface while still containing no evidence.

33 Live-But-Unpopulated Artifacts

Section status. `protocol_defined`.

Stage3-6 instantiates a new artifact class: live-but-unpopulated operational records. These include session-zero manifests, live collection manifests, empty collection logs, live dataset manifests, live reproducibility manifests, live decision-record stubs, and live evidence-insertion registers.

Each such surface must explicitly preserve:

State	Meaning
`schema_defined`	Field grammar exists, but no operational object family is yet instantiated.
`contract_defined`	Allowed relation among objects exists, but no package is assembled.
`package_ready`	The package grammar exists and may be evaluated for completeness.
`instantiated_empty`	The object family exists locally with null-bearing or empty-bearing fields only.
`frozen_local_unfiled`	Local freeze semantics exist, but no external filing is claimed.
`rehearsal_ready`	Rehearsal grammar and rehearsal-kit checks pass, but no live execution shell exists yet.
`live_artifacts_instantiated_unpopulated`	Live operational shells exist, but every collection-facing structure remains empty.
`session_zero_ready`	Session-zero initialization is structurally ready, but still not started.
`collection_openable_under_controls`	Pre-collection admissibility checks pass under contamination and abort controls.
`collection_open_pre_evidential`	Controlled execution may open as a live state without any populated evidence.
`stage3_7_first_collection_openable`	Pre-runtime-health and clean-room audits pass, the opening gate is structurally open, but no collection-start event has been executed.
`collection_started`	A real authorized collection-start event exists.
`evidence_generated`	Real dataset population or admissible log events exist.
`adjudication_result_recorded`	Reviewer-governed or adjudicator-governed decision records exist.
`paper10_final_authorized`	A later separate manuscript-authorization layer has passed.
`dry_run_open`	Dry-run structure may execute without evidence or adjudication.
`externally_filed`	A real external filing reference exists.
`populated_with_admissible_evidence`	The object family is populated by real admissible evidence.
`adjudication_result_bearing`	Reviewer and adjudicator outputs exist and may support result-bearing sections.

- `collection_not_started`;
- `evidence_generation_absent`;
- `adjudication_not_open`;
- explicit empty markers such as `empty_dataset = true` where applicable.

Therefore live artifact instantiation is not population, and live shells must never be treated as if they were evidence-bearing records.

34 Session-Zero Protocol

Section status. `protocol_defined`.

Session-zero is the bounded initialization checkpoint that binds local frozen record routing, live collection manifests, empty dataset manifests, empty collection logs, empty decision stubs,

contamination controls, abort controls, and the operator checklist. Session-zero may become ready without collection starting.

Session-zero remains distinct from:

- collection start;
- dataset population;
- reviewer action;
- adjudicator action.

35 Pre-Collection Admissibility Gate

Section status. `protocol_defined`.

The pre-collection gate asks whether collection may be opened under controls, not whether collection has already started. This gate checks session-zero readiness, emptiness of live artifacts, emptiness of collection logs, emptiness of decision stubs, absence of evidence, absence of adjudication, and absence of detected contamination.

Passing this gate yields only `collection_openable_under_controls`. It does not yield `collection_started`.

36 Contamination, Abort, and Invalidation Controls

Section status. `protocol_defined`.

Stage3-6 adds a more operational invalidation grammar:

- **contamination event**: any unauthorized dataset population, evidence-bearing log entry, or decision-bearing record.
- **abort event**: any operator-controlled termination of the controlled opening before collection start.
- **invalidated session**: any session whose pre-collection boundary, blinding route, or provenance cleanliness is breached.

These control classes are meaningful only because Stage3-6 now contains live artifacts. Without them, contamination and abort would remain abstract policy tokens rather than execution-opening constraints.

37 Operator Checklist

Section status. `protocol_defined`.

The operator checklist now matters as a transition object. It binds the local frozen record route, live manifest route, empty-log route, empty-dataset state, contamination rules, reset rules, and explicit confirmation that no evidence exists. Until it is complete, Stage3-6 must not move from `session_zero_ready` to `collection_openable_under_controls`.

38 Collection-Start Boundary

Section status. `formalized`.

`collection_open_pre_evidential` is not `collection_started`. The boundary is crossed only by a later real operator-authorized collection start event with provenance, contamination-free status, and a first admissible log event.

39 Evidence-Generation Boundary

Section status. `formalized`.

Evidence generation begins only when a real admissible population event enters the human comparator dataset family, matched intervention family, negative-control family, collection log, reproducibility package, or decision-linked provenance chain. No Stage3-6 artifact emitted in this pass crosses that boundary.

40 Decision-Record Boundary

Section status. `formalized`.

The live decision-record stub is an execution-opening shell only. It is not a decision, not an adjudicator action, not a reviewer action, and not an outcome. The manuscript must preserve that distinction because a populated decision record belongs to a later evidential and adjudicative state family.

41 Transition from Session-Zero to First Admissible Collection

Section status. `blocked_until_execution`.

The next real transition is now sharper than in earlier passes:

1. session-zero becomes ready;
2. the pre-collection gate opens collection under controls;
3. Stage3-6 opens a live but still pre-evidential state;
4. a later explicit collection-start command instantiates the first admissible collection event;
5. only then may the first dataset-population event exist.

42 Runtime Health Audit

Section status. `protocol_defined`.

Stage3-6B adds a runtime-health audit that asks whether the repository is actually executable on a bounded local environment rather than only formally specified. The required audit surfaces include:

- toolchain visibility;
- dependency-install behavior under the declared repository install route;

- repository build behavior;
- repository test behavior;
- bounded local-start behavior;
- verifier-chain behavior for source-of-truth, comparative closure, Stage3-6, Stage3-5, and Paper 9 handoff.

The runtime-health audit may pass with warnings. Such warnings may include engine mismatch, fallback-install compatibility, or dependency advisories. A warning-bearing pass is still operationally meaningful, but it is not epistemically elevated into evidence.

43 Clean-Room Reproducibility

Section status. `protocol_defined`.

Stage3-6B also adds a clean-room reproducibility audit. The clean-room audit asks whether a bounded local copy can reinstall dependencies, rebuild the repository, rerun key verifiers, and regenerate Stage 3 artifacts without relying on hidden machine-specific state. Clean-room reproducibility may pass with warnings when a compatibility route is required, provided the route is explicitly documented and the resulting build remains honest.

44 Dependency and Build Assumptions

Section status. `formalized`.

The runtime-health layer now makes dependency and build assumptions explicit:

- the declared Node engine remains part of the reproducibility contract;
- the preferred install route remains `npm ci`;
- a fallback install route may be recorded only as a compatibility surface;
- build success must remain distinct from evidence generation;
- test success must remain distinct from adjudication.

45 Reproducible Artifact Regeneration

Section status. `protocol_defined`.

Stage3-6B adds a source-driven artifact-regeneration audit. This audit checks whether tracked artifacts can be regenerated from the current source tree and whether the resulting diffs are:

- absent;
- provenance-only;
- semantically stale but honestly fixed; or
- blocking.

This matters because Stage 3 now depends on a growing artifact family. The audit therefore separates semantic drift from ordinary lineage and timestamp drift, and it refuses to treat manual cosmetic rewriting as acceptable regeneration.

46 Local Path Leakage Policy

Section status. `formalized`.

Clean-room readiness now requires that live normative surfaces remain free of machine-specific absolute paths. Historical planning notes or preserved snapshots may still contain old local references, but they must remain explicitly classified as historical rather than live normative dependencies.

47 Controlled Execution Opening Audit

Section status. `protocol_defined`.

The controlled-execution opening layer is now itself audited. Stage3-6B does not replace Stage3-6; it checks whether Stage3-6 is actually executable, rebuildable, and reproducible. Therefore the live-but-unpopulated artifacts, session-zero gate, and pre-collection gate must now survive both runtime health verification and clean-room reconstruction.

48 Session-Zero Reproducibility

Section status. `protocol_defined`.

Session-zero reproducibility means that a rebuilt local tree can recreate the session-zero manifest, operator checklist state, and failure-mode assessment without any hidden local inputs and without producing collection-start or evidence-bearing records.

49 Pre-Collection Gate Reproducibility

Section status. `protocol_defined`.

Pre-collection gate reproducibility means that a rebuilt local tree can recreate the pre-collection admissibility result, contamination checklist, and invalidation status while still preserving:

- `collection_not_started`;
- `evidence_generation_absent`;
- `adjudication_not_open`.

50 Why Reproducibility Does Not Imply Evidence

Section status. `formalized`.

Reproducibility shows that the formal and executable surfaces are rebuildable. It does not show that any dataset has been populated, that any human evidence exists, that any reviewer or adjudicator has acted, or that any result-bearing section is now licensed.

51 Why Runtime Health Does Not Imply Adjudication

Section status. `formalized`.

Runtime health shows that the code path and verifier path are alive. It does not imply that collection has started, that evidence has been generated, that reviewer-governed interpretation

exists, or that adjudication is open. A healthy runtime is still a pre-evidential state until later real execution events occur.

52 Transition Conditions for First Admissible Collection

Section status. `protocol_defined`.

Stage3-7 may be classified as `stage3_7_first_collection_openable` only when runtime-health and clean-room reproducibility both pass honestly, path leakage remains clear on live normative surfaces, artifact regeneration is non-blocking, Stage3-6 remains openable, and Paper 10 remains non-final and non-result-bearing.

Even then, the current state must still preserve:

- `collection_start_not_executed`;
- `evidence_generation_absent`;
- `adjudication_not_open`;
- `decision_record_absent`.

53 Claim Grammar for Eventual Execution-Era Statements

Section status. `protocol_defined`.

Any later admissible result-bearing statement must reference:

- preregistration identifier;
- freeze identifier;
- dataset family identifier;
- negative-control family identifier;
- analysis shell identifier;
- reviewer package identifier;
- adjudicator package or decision identifier.

Forbidden grammar remains explicit:

- no leap from comparator similarity to human equivalence;
- no leap from transport-sensitive comparison to identity preservation;
- no leap from admissibility to metaphysical subjecthood;
- no leap from dry-run success to public-release recommendation.

54 Theorem, Proposition, and Protocol Mapping

Section status. `formalized`.

Family	Dependency set	Formalized now	Requires execution	Remains forbidden
Comparator channel contract	BaseCore, BPF, Stage 2, Stage3-1	yes	no	result claim
Freeze architecture	Stage3-1, Stage3-2, Stage3-3	yes	no	external filing claim
Reviewer package architecture	Stage3-2, Stage3-3	yes	no	review outcome
Pilot package architecture	Stage3-3	yes	no	evidence claim
Dry-run opening grammar	Stage3-3, Stage3-4	yes	no	adjudication claim
Human comparator result section	all above + admissible data	no	yes	human equivalence
Adjudication outcome section	all above + reviewer/adjudicator layer + admissible evidence	no	yes	metaphysical closure

55 What Paper 10 Could and Could Not Change

Section status. `formalized`.

Paper 10 is the only layer in the present corpus designed to move evidence out of the purely internal-support regime. Even a successful execution would not authorize all possible interpretations.

Claim family	Possible post-execution movement	Still forbidden by default
Runtime reproducibility	from local/internal to externally reproduced under protocol	consciousness, subjecthood, phenomenality
Human comparator similarity	from unpopulated protocol to measured channel comparison	human equivalence or moral parity
Bridge predicate discrimination	from BPF scaffold/internal surfaces to adjudicated comparative evidence	phenomenal experience simpliciter
Negative-control survival	from internal controls to externally witnessed controls	theorem confirmation beyond stated assumptions
Failure or inconclusive result	from open uncertainty to documented non-support or bounded inconclusiveness	rhetorical rescue by post hoc reinterpretation

Proposition 55.1 (External evidence is not interpretive closure)**.** *Executed Paper 10 evidence, even if admissible, would not by itself establish human equivalence, metaphysical subjecthood, biological life, phenomenal experience, or moral status.*

Proof. The Paper 10 protocol measures channel comparison, adjudicator behavior, negative controls, provenance, contamination, and statistical distinguishability under frozen rules. None of those objects is identical to the listed interpretive predicates. Moving a claim from internal support to external evidence therefore changes its evidential status, not its semantic type. □

56 Section-Status Ledger

Section status. `formalized`.

Section	Status class	Unlock condition
Dependency stack	formalized	Update only if upstream stack changes
Release-vs-local boundary	formalized	Update only if release authority changes
Lineage semantics	formalized	Keep historical-vs-live distinction intact
Frozen preregistration architecture	protocol_defined	Actual local lock still absent
Reviewer candidate architecture	protocol_defined	Selection not started
Blinding assignment logic	protocol_defined	Blinding package not instantiated
Execution pilot package architecture	protocol_defined	Pilot remains unexecuted
Reproducibility skeleton architecture	protocol_defined	Package unpopulated
Dataset population boundary	formalized	Keep defined-vs-populated split intact
Adjudication decision grammar	protocol_defined	Decision tokens remain uninstantiated
Future result sections	blocked_until_execution	Requires admissible execution-era artifacts
External adjudication discussion	blocked_until_external_adjudication	Requires admissible reviewer-governed evidence
Human equivalence / superiority / metaphysical consequences	forbidden_claim_surface	Remains outside current authority surface

57 Evidence Insertion Map

Section status. `evidence_pending`.

Human comparator result insertion point. Admissible evidence required: frozen preregistration record, typed human comparator dataset, negative-control bundle, statistical shell output, reviewer-selection package, reviewer report family.

External adjudication result insertion point. Admissible evidence required: reviewer report set, adjudicator decision record, blinding log, conflict disclosures, reproducibility package, failure and inconclusive pathway disclosures.

Transport-sensitive comparison insertion point. Admissible evidence required: source-target pairing manifest, transition lineage map, axis-specific error budgets, transport negative controls, adjudicator layer.

Instantiated-empty to evidential transition point. Admissible evidence required: local frozen record upgraded by real lock event, real candidate population, real assignment event, real collection session, real decision record population, and real reproducibility package population.

58 Open Burdens and Blocked Sections

Section status. `blocked_until_execution`.

`blocked_until_execution`. This section requires executed campaigns with human participants, frozen external filing records, populated admissible datasets, actual reviewer and adjudicator assignments, reviewer-governed evidence assessment, real collection-start events, and later adjudicator-layer completion. It contains no results and must remain non-result-bearing until those artifacts exist.

Open burdens remain explicit:

- real local frozen preregistration lock absent;
- external filing absent;
- reviewer selection not started;
- adjudicator assignment not started;
- actual blinding assignment absent;
- admissible human comparator dataset absent;
- admissible matched-intervention dataset absent;
- admissible negative-control dataset absent;
- collection sessions not started;
- rehearsal event log not started;
- reproducibility package unpopulated;
- reviewer and adjudicator decision records absent.

59 Execution-Era Unlock Conditions

Section status. `protocol_defined`.

Before any result-bearing section may become writable, the following must all be true:

1. a frozen preregistration record exists;
2. any required external filing state is real and documented;
3. candidate pools are populated by real identities with provenance;
4. reviewer selection is instantiated;
5. adjudicator assignment is instantiated;

6. blinding assignment is instantiated;
7. admissible datasets and control bundles are populated;
8. collection logs and execution sessions are populated by real events;
9. reproducibility package slots are populated;
10. decision grammar is instantiated by real evidence.

60 Stronger Execution-to-Result Transition Map

Section status. `blocked_until_execution`.

The transition from current Stage3-6 controlled opening to a result-bearing paper now has six distinct unlock layers:

1. **Instantiation layer**: empty records, pools, stubs, and rehearsal kit exist.
2. **Controlled opening layer**: live operational shells, session-zero, and pre-collection controls exist while collection still remains not started.
3. **Population layer**: real humans, real sessions, and real evidence populate those empty structures.
4. **Admissibility layer**: controls, provenance, blinding, and declared shells survive audit.
5. **Decision layer**: reviewer and adjudicator records become populated by real independent action.
6. **Manuscript layer**: only then may result-bearing Paper 10 sections be written.

61 Why Paper 10 Still Cannot Become Final Yet

Section status. `formalized`.

Paper 10 cannot become final because the current repo still lacks the evidence bearing surfaces that would convert protocol architecture into scientific claims. The manuscript may now specify the machinery of later freezing, selection, packaging, dry-run, provenance closure, instantiated-empty records, and rehearsal-facing governance. It still may not write conclusions whose truth depends on evidence that does not yet exist.

62 Future Execution Pathway to Real Stage 3

Section status. `blocked_until_execution`.

The next executable pathway is not "publish Paper 10." It is:

1. populate the local frozen record honestly;
2. populate reviewer and adjudicator pools honestly;
3. instantiate actual reviewer, adjudicator, and blinding assignments;

4. validate dry-run, rehearsal, and controlled-opening integrity;
5. issue a real collection-start event under Stage3-6 controls;
6. populate admissible datasets, collection logs, and control bundles;
7. run declared statistical shells;
8. bind reviewer and adjudicator outputs;
9. only then write result-bearing sections.

63 Boundary to Stage 4 Metaphysical Consequence Analysis

Section status. `forbidden_claim_surface`.

Stage 4 remains outside the authority surface of this document. Even later successful execution-era evidence would still not automatically license metaphysical subjecthood, moral status, or cross-substrate identity preservation.

64 Appendices of Schemas and Templates

Section status. `protocol_defined`.

Appendix A: Frozen Preregistration Record Template

Section status. `protocol_defined`. Primary reference: `docs/frozen_preregistration_record_schema.v1.js`
Admissible evidence required before population: actual local lock and, where required by protocol, real external filing reference.

Appendix B: Human Comparator Preregistration Template

Section status. `protocol_defined`. Primary reference: `docs/human_comparator_preregistration_template.`
Admissible evidence required before population: channel-level freeze and later dataset lineage.

Appendix C: Reviewer Candidate Schema

Section status. `protocol_defined`. Primary reference: `docs/reviewer_candidate_schema.v1.json`.
Admissible evidence required before population: real candidate package and conflict disclosures.

Appendix D: Adjudicator Candidate Schema

Section status. `protocol_defined`. Primary reference: `docs/adjudicator_candidate_schema.v1.json`.
Admissible evidence required before population: real assignment package and decision-authority boundary.

Appendix E: Blinding Assignment Template

Section status. `protocol_defined`. Primary reference: `docs/blinding_assignment_manifest.v1.json`.
Admissible evidence required before population: authorized assignment route and later unblinding event if one exists.

Appendix F: Execution Pilot Package Manifest

Section status. protocol_defined. Primary reference: docs/execution_pilot_package_manifest.v1.json.
Admissible evidence required before population: real session manifests and later populated bundle references.

Appendix G: Dataset Contract Summary

Section status. protocol_defined. Primary references: docs/matched_intervention_dataset_contracts.v1
and docs/negative_control_dataset_contracts.v1.json.
Admissible evidence required before result use: populated admissible datasets.

Appendix H: Negative Control Bundle Summary

Section status. protocol_defined. Primary reference: docs/pilot_negative_control_bundle.v1.json.
Admissible evidence required before result use: populated bundle and control admissibility record.

Appendix I: Statistical Shell Summary

Section status. protocol_defined. Primary reference: docs/statistical_analysis_shells.v1.json.
Admissible evidence required before result use: shell execution output on real admissible inputs.

Appendix J: Adjudication Decision Grammar Summary

Section status. protocol_defined. Decision tokens retained as grammar only: externally_admissible_but_i
protocol_invalidated, adjudication_failed, adjudication_passed, reviewer_consensus_absent,
adjudicator_decision_withheld.

Appendix K: Local Frozen Record Example

Section status. protocol_defined. Primary reference: docs/local_frozen_preregistration_record_templa
Admissible evidence required before result use: real local lock event and, where required by route, external filing provenance.

Appendix L: Reviewer Candidate Pool Template

Section status. protocol_defined. Primary reference: docs/reviewer_candidate_pool_template.v1.json.
Admissible evidence required before result use: real reviewer identities, conflict disclosures, and selection events.

Appendix M: Adjudicator Candidate Pool Template

Section status. protocol_defined. Primary reference: docs/adjudicator_candidate_pool_template.v1.jso
Admissible evidence required before result use: real adjudicator identity, assignment provenance, and decision-authority declaration.

Appendix N: Blinding Assignment Stub

Section status. `protocol_defined`. Primary references: `docs/blinding_assignment_stub.v1.json` and `docs/reviewer_blinding_stub.v1.json`.
Admissible evidence required before result use: real assignment route and real unblinding trace if one later exists.

Appendix O: Execution Session Template

Section status. `protocol_defined`. Primary reference: `docs/execution_session_template.v1.json`.
Admissible evidence required before result use: real collection session and real session lineage.

Appendix P: Empty Collection Log Schema

Section status. `protocol_defined`. Primary reference: `docs/collection_log_schema.v1.json`.
Admissible evidence required before result use: populated log entries from real collection events.

Appendix Q: Empty Decision Record Template

Section status. `protocol_defined`. Primary reference: `docs/decision_record_template.v1.json`.
Admissible evidence required before result use: real reviewer and adjudicator action, not grammar only.

Appendix R: Reproducibility Manifest Example

Section status. `protocol_defined`. Primary reference: `docs/reproducibility_package_manifest_empty.v1.`
Admissible evidence required before result use: populated package slots, dataset references, shell outputs, and reviewer-facing bundles.

Appendix S: Evidence Insertion Manifest

Section status. `protocol_defined`. Primary reference: `docs/evidence_insertion_manifest.v1.json`.
Admissible evidence required before result use: actual artifact refs, actual decision refs, and populated provenance chains.

Appendix T: Rehearsal-Kit Map

Section status. `protocol_defined`. Primary references: `docs/execution_pilot_rehearsal_kit.v1.json` and `docs/dry_run_sequence_plan.v1.json`.
Admissible evidence required before result use: none at rehearsal time, but no rehearsal artifact may be promoted into evidence.

Appendix U: Dry-Run Event Log Grammar

Section status. `protocol_defined`. Primary reference: `docs/rehearsal_event_log_schema.v1.json`.
Allowed event classes remain rehearsal-only and do not imply human data, reviewer action, or adjudicator action.

Appendix V: Controlled Execution Opening Policy

Section status. protocol_defined. Primary reference: docs/controlled_execution_opening_policy.v1.jso
Admissible evidence required before result use: none at opening time, but no controlled-opening artifact may be promoted into evidence by itself.

Appendix W: Session-Zero Protocol and Checklist

Section status. protocol_defined. Primary references: docs/session_zero_protocol.v1.json and docs/session_zero_operator_checklist.v1.json.
Admissible evidence required before result use: a later real collection-start event and later populated log entries.

Appendix X: Pre-Collection Gate and Contamination Controls

Section status. protocol_defined. Primary references: docs/pre_collection_admissibility_gate.v1.json docs/pre_collection_gate_criteria.v1.json, and docs/pre_collection_contamination_checklist.v1.jso
Admissible evidence required before result use: a later contamination-free collection-start event and later admissible dataset population.

Appendix Y: Live-But-Unpopulated Artifact Map

Section status. protocol_defined. Primary references: artifacts/external_adjudication/stage3_6/live_ artifacts/external_adjudication/stage3_6/live_human_comparator_dataset_manifest.json, and parallel Stage3-6 live manifests.
Allowed interpretation: live operational shell only. Forbidden interpretation: populated dataset, evidence, or adjudication.

Appendix Z: First Admissible Collection Unlock Map

Section status. protocol_defined. Primary references: docs/first_admissible_collection_unlock_map.v1 and docs/FIRST_ADMISSIBLE_COLLECTION_UNLOCK_MAP.md.
Admissible evidence required before result use: actual reviewer selection, actual adjudicator assignment, actual blinding assignment, operator authorization, collection-start command, and first populated collection-log event.

Appendix AA: Runtime Health Audit

Section status. protocol_defined. Primary references: docs/runtime_health_audit.v1.json, docs/RUNTIME_HEALTH_AUDIT.md, and artifacts/runtime_health/runtime_health_audit.json.
Allowed interpretation: executable health and verifier-chain status only. Forbidden interpretation: evidence, collection start, or adjudication.

Appendix AB: Clean-Room Reproducibility Audit

Section status. protocol_defined. Primary references: docs/clean_room_reproducibility_audit.v1.json, docs/CLEAN_ROOM_REPRODUCIBILITY_AUDIT.md, and artifacts/runtime_health/clean_room_reproducibility
Allowed interpretation: bounded rebuildability only. Forbidden interpretation: external adjudicative support, empirical support, or manuscript finality.

Appendix AC: Local Path Leakage Audit

Section status. `protocol_defined`. Primary references: `docs/PATH_AND_LOCAL_LEAKAGE_AUDIT.md` and `artifacts/runtime_health/path_and_local_leakage_audit.json`.
Allowed interpretation: leak classification only. Forbidden interpretation: claim that historical snapshots were rewritten or erased.

Appendix AD: Artifact Regeneration Audit

Section status. `protocol_defined`. Primary references: `docs/ARTIFACT_REGENERATION_AUDIT.md` and `artifacts/runtime_health/artifact_regeneration_audit.json`.
Allowed interpretation: regeneration and drift classification only. Forbidden interpretation: result-bearing evidence insertion.

Appendix AE: Stage3-7 First Collection Opening Criteria

Section status. `protocol_defined`. Primary references: `docs/stage3_7_first_admissible_collection_open` `docs/first_admissible_collection_start_policy.v1.json`, `artifacts/external_adjudication/stage3_7/s` and `artifacts/external_adjudication/stage3_7/first_collection_opening_manifest.json`.
Allowed interpretation: first-collection opening criteria only. Forbidden interpretation: executed collection start, evidence generation, or adjudication.

65 Blocked Result Surfaces

Section status. `blocked_until_external_adjudication`.

65.1 Human Comparator Results

`blocked_until_execution.` This section requires executed campaigns with human participants, frozen external filing records, populated admissible datasets, actual reviewer and adjudicator assignments, reviewer-governed evidence assessment, real collection-start events, and later adjudicator-layer completion. It contains no results and must remain non-result-bearing until those artifacts exist.

Admissible evidence required. Frozen preregistration record required. Artifact references required. Negative controls required. Statistical shell required. Reviewer package required.

65.2 External Adjudication Results

`blocked_until_execution.` This section requires executed campaigns with human participants, frozen external filing records, populated admissible datasets, actual reviewer and adjudicator assignments, reviewer-governed evidence assessment, real collection-start events, and later adjudicator-layer completion. It contains no results and must remain non-result-bearing until those artifacts exist.

Admissible evidence required. Independent reviewer reports required. Conflict disclosures required. Blinding log required. Adjudicator decision record required.

65.3 Comparative Consequence Claims

`blocked_until_execution.` This section requires executed campaigns with human participants, frozen external filing records, populated admissible datasets, actual reviewer and adjudicator assignments, reviewer-governed evidence assessment, real collection-start events, and later adjudicator-layer completion. It contains no results and must remain non-result-bearing until those artifacts exist.

Admissible evidence required. Execution-era artifacts required. Adjudication-facing evidence required. Identity-preservation, metaphysical, and moral-status claims remain forbidden even after this section becomes writable.

66 Non-Claim Ledger

Section status. `formalized`.

This manuscript does not establish consciousness, phenomenality in the strong sense, human equivalence, superiority, inferiority, metaphysical subjecthood, cross-substrate identity preservation, or moral status.